Chapter Seven

Set

The word "Set" is familiar to us. For example: tea set, sofa set, dinner set, set of books etc. German mathematician George Cantor (1845-1918) explained the idea about set. His explanation of set is known as Set Theory in mathematics. Starting from the elementary concepts of set through symbols and figures we need to acquire knowledge about set. In this chapter, different types, operations and properties of sets have been discussed.

At the end of the chapter, the students will be able to–

- explain and form sets;
- explain finite set, universal set, complementary set, empty set, disjoint set and express the formation of these sets by symbols;
- explain the formation of union and intersections of sets;
- verify and prove simple properties of set operations by Venn diagram and examples;
- solve problems by applying the properties of set.

7·1 Set

A well-defined collection of objects of real or imaginative world is called set. Examples of sets of well-defined objects are: the first five English alphabets, the countries of Asia, natural numbers etc. It is to be determined particularly which object is included in the set under consideration and which is not. There is no repetition and order of the objects in set.

Each object of a set is called an element of the set. Set is generally denoted by capital letters of English alphabet as A, B, C,... X, Y, Z and the elements are expressed in small letters as a, b, c,... x, y, z.

The set is expressed by the symbol { } which includes the elements of the set. For example : the set of a, b, c is { a, b, c}. The set of the Tista, the Meghna, the Jamuna and the Brahmaputra rivers is {Tista, Megna, Jamuna, Brahmaputra}. The set of the first two even natural numbers is { 2, 4} ; the set of the factors of 6 is {1, 2, 3, 6} etc.

Let x be an element of set A. Mathematically, it is expressed by $x \in A$.
$x \in A$ is read x is an the element of the set A, (*x belongs to A)*. For example, if $B = \{m, n\}$, then $m \in B$ and $n \in B$.

Example 1. If the set of the first five odd numbers is A, $A = \{1, 3, 5, 7, 9\}$

Activity :
1. Write a set of SAARC countries.
2. Write a set of prime numbers from 1 to 20.
3. Write a set of any four numbers between 300 and 400 which are divisible by 3.

7·2 Methods of expressing set

Set can be expressed mainly in two methods : (1) Tabular Method
(2) Set Builder Method.

(1) Tabular Method : In this method, all the elements of a set are mentioned particularly by enclosing them in second brackets { } and if there is more than one element, the elements are separated by using comma(,). For example :

A= {1, 2, 3}, B = {x, y, z}, C = {100}, D = {Rose, Tube-rose}, E = {Rahim, Sumon, Suvro, Changpai} etc.

(2) Set Builder Method : In this method, conditions are given to determine the elements of sets without mentioning them particularly. For example, if the set of natural even numbers which are smaller than 10 is A, A= {x: x natural even number, x < 10}. Here":" means "such that" or in brief "such". In set builder method, one unknown quantity or variable is placed before ":" sign inside { } and then required conditions are applied to the variable. For example, let us express the set {3, 6, 9, 12} in set builder method. Observe that 3, 6, 9, 12 are natural numbers which are divisible by 3 and are not greater than 12. In this case, if the element of the set is considered to be the variable y. the conditions that will be applied are : y is a natural number, multiple of 3 and not more than 12 ($y \leq 12$).

Therefore, in set builder method, it will be { y : y is a natural number, multiple of 3 and $y \leq 12$ } .

Example 2. Express the set $P = \{4, 8, 12, 16, 20\}$ in set builder method.

Solution : The elements of the set P are 4, 8, 12, 16, 20.
Here, each element is an even number, multiple of 4 and not greater than 20.
$\therefore$ $P = \{x : x$ is x even number, multiple of 4 and $x \leq 20\}$

Example 3. Express the set $Q = \{x : x$ are all the factors of $42\}$ in Tabular method.

Solution : Q is the set of factors of 42

Here, $42 = 1 \times 42 = 2 \times 21 = 3 \times 14 = 6 \times 7$

$\therefore$ The factors of 42 are $1, 2, 3, 6, 7, 14, 21, 42$.

Required set $Q = \{1, 2, 3, 6, 7, 14, 21, 42\}$

Activity :

1. Express the set $A = \{3, 6, 9, 12, 15, 18\}$ in set builder method.
2. Express the set $B = \{x : x$ is a factor of $24\}$ in tabular method.

7·3 Classification of Sets

Finite Set :

If the number of elements of a set can be determined by counting, it is called a finite set. For example: $A = \{a, b, c, d\}$, $B = \{5, 10, 15, ..., 100\}$ etc. are finite sets. Here, there are 4 elements in set A and 20 elements in set B.

Infinite Set :

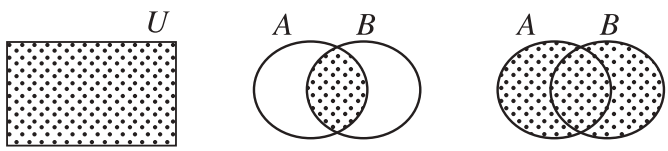
The set whose number of elements can not be determined by counting is called an infinite set. One example of infinite set is, a set of natural numbers, $N = \{1, 2, 3, 4...\}$. Here, the number of elements of set N is innumberable, which can not be determined. In this chapter, only the finite sets will be discussed.

Empty Set :

The set which has no element is called an empty set. An empty set is expressed by the symbol Ø.

7·4 Venn-diagram

John Venn (1834-1883) introduced the method of expressing sets by diagrams. These diagrams are named after Venn and called Venn diagram. Generally, in Venn-diagram, rectangular and circular regions are used. Venn-diagrams of some sets are shown below :



By using Venn-diagrams, the properties of sets and operations on sets can be easily determined.

7.5 Subset

Let, $A = \{a, b\}$ is a set. With the elements of the set A, we can form the sets $\{a, b\}, \{a\}, \{b\}$. The sets $\{a, b\}, \{a\}, \{b\}$ are the subsets of A.

As many sets are formed from the elements of a set, each of those sets is a subset of the given set.

For example : if $P = \{2, 3, 4, 5\}$ and $Q = \{3, 5\}$, the set Q is the subset of P. That means, Q $\subseteq$ P because the elements 3, 5, of set Q are also included in set P. Subsets are expressed by using '$\subseteq$' symbol.

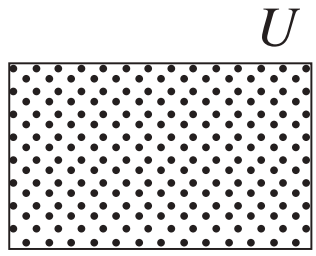
Example 4 : Write the subsets of the set $A = \{1, 2, 3\}$.

Solution : The subsets of the set A are shown below :

$\{1, 2, 3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1\}, \{2\}, \{3\}$, Ø.

Universal Set :

If all the sets in the discussion are the subsets of a particular set, that particular set is called the universal set. A universal set is expressed by the symbol U. For example, in a school, the set of all the students is a universal set and the set of the students of class eight is the subset of the universal set.



All sets are subsets of the universal set in a given context.

Example 5. If $A = \{1, 2, 3, 4, 5, 6\}, B = \{1, 3, 5\}, C = \{3, 4, 5, 6\}$ determine the universal set.

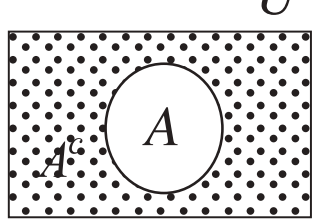
Solution : Given, $A = \{1, 2, 3, 4, 5, 6\}, B = \{1, 3, 5\}, C = \{3, 4, 5, 6\}$.

Here, the elements of set B are 1, 3, 5 and the elements of C are 3, 4, 5 that are included in set A.

$\therefore$ in respect of these two sets B and C, the universal set is A.

Complement of a Set :

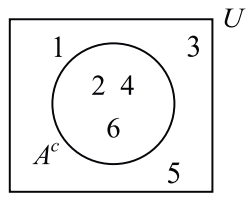
If U is an universal set and set A is the subset of U, then the set of all elements that are excluded from set A, is called the complement set of set A. The complement set of A is denoted by A^c or A'.



Suppose, in class eight, out of 60 students, 9 students are absent. If the set of all the students of class eight is considered as a universal set, the set of those 9 absent students will be the compliment set of the set of (60 - 9) or 51 present students.

Example 6. If $U = \{1, 2, 3, 4, 5, 6\}$ and $A = \{2, 4, 6\}$, determine A^c.

Solution : Given, $U = \{1, 2, 3, 4, 5, 6\}$ and $A = \{2, 4, 6\}$.



$\therefore$ A^c = The complement set of A
= The set of the elements excluding the elements of A
= $\{1, 3, 5\}$

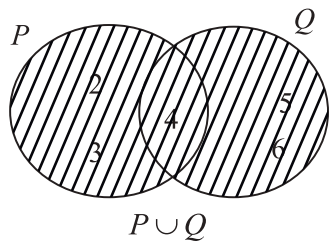
Required set $A^c = \{1, 3, 5\}$

Activity : If $A = \{a, b, c\}$, find the subsets of A and find the complementary set of any three of them.

7.6 Set operations

Union of sets

Let P=$\{2, 3, 4\}$ and $\{Q = 4, 5, 6\}$ be two sets. Here, all the elements included in sets P and Q are $2, 3, 4, 5, 6$. The set formed by all the elements of sets P and Q is $\{2, 3, 4, 5, 6\}$, which is the union set of P and Q.



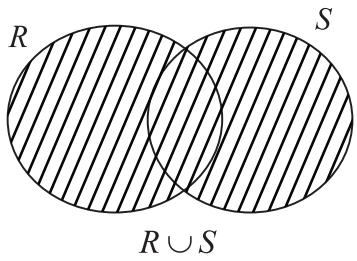
The set formed by all the elements of two or more sets is called a union set. Let, A and B be two sets. The set formed by all the elements of the sets A and B is expressed as $A \cup B$ and read as 'A union B'.
In set builder method, $A \cup B = \{x : x \in A$ or $x \in B\}$

Example 7. C = {Razzaq, Sakib, Alok} and D = {Alok, Mushfiq}, then find $C \cup D$.

Solution : Given that, C = {Razzaq, Sakib, Alok} and D = {Alok, Moshfiq}
$\therefore$ $C \cup D$ = {Razzaq, Sakib, Alok} $\cup$ {Alok, Mushfiq}
= {Razzaq, Sakib, Alok, Mushfiq}

Example 8. $R = \{x : x$ is factor of $6\}$ and $S = \{x : x$ is factor of $8\}$, then find $R \cup S$.

Solution : Given that, $R = \{x : x \text{ is factor of } 6\}$
$= \{1, 2, 3, 6\}$
and $S = \{x : x \text{ is factor of } 8\}$
$= \{1, 2, 4, 8\}$
$\therefore\ R \cup S = \{1, 2, 3, 6\} \cup \{1, 2, 4, 8\}$
$= \{1, 2, 3, 4, 6, 8\}$



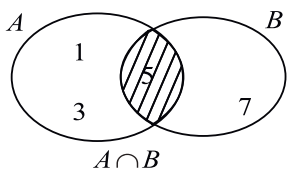
Intersection of sets

Suppose, Rina can read and write both Bangla and Arabic and Joya can read and write both Bangla and Hindi. The set of the languages that Rina can read and write is {Bangla, Arabic}, the set of the languages that Joya can read and write is {Bangla, Hindi}. We can see, that the language that both Rina and Joya can read and write is Bangla and the set of it is {Bangla}. Here, the set {Bangla} is the intersection of the two sets.

The set formed by the common elements of two or more sets is called intersection of sets.

Let, A and B are two sets. The intersection of the sets A and B is denoted by $A \cap B$ and read 'A intersection B'. In the set builder notation the set is $A \cap B = \{x : x \in A \text{ and } x \in B\}$

Example 9. If $A = \{1, 3, 5\}$ and $B = \{5, 7\}$, find $A \cap B$.



Solution : Given that, $A = \{1, 3, 5\}$ and $B = \{5, 7\}$
$\therefore\ A \cap B = \{1, 3, 5\} \cap \{5, 7\} = \{5\}$

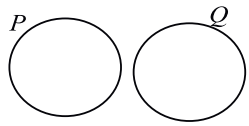
Example 10. If $P = \{x : x \text{ is the multiple of 2 and } x \le 8\}$ and $Q = \{x : x \text{ is the multiple of 4 and } x \le 12\}$, find $P \cap Q$.

Solution : Given that, $P = \{x : x \text{ is the multiple of 2 and } x \le 8\}$
$= \{2, 4, 6, 8\}$
and $Q = \{x : x \text{ is the multiple of 4, } x \le 12\}$
$= \{4, 8, 12\}$
$\therefore\ p \cap Q = \{2, 4, 6, 8\} \cap \{4, 8, 12\} = \{4, 8\}$

Activity : $U = \{1, 2, 3, 4\}, A = \{1, 2, 3\}, B = \{2, 3, 4\}, C = \{1, 3\}$
show the sets $U \cap A$, $C \cap A$ and $B \cup C$ in Venn diagram.

Disjoint Set

Let, there are two villages side-by-side in Bangladesh. The farmers of one village grow paddy and jute and the farmers of the other village grow potato and vegetables in their fields. If we consider two sets of the cultivated crops, we get {paddy, jute} and {potato, vegetables}. There is no common crop between the two sets. That means, the farmers of two villages do not grow the same crops. Here, the two sets are disjoint sets to each other.



If there is no common element between the elements of two sets, the sets are called disjoint sets.

Let, , A and B are two sets, A and B will be disjoint sets to each other if $A \cap B = \emptyset$.

If the intersection of two sets is an empty set, they are disjoint sets.

Example 11. $A = \{x : x$ is odd natural number and $1 < x < 7\}$ and $B = \{x : x$ is factor of $8\}$, then show that the sets A and B are disjoint.

Solution : Given that, $A = \{x : x$ is odd natural number and $1 < x < 7\}$

$= \{3, 5\}$

and $B = \{x, x$ is factor of $8\}$

$= \{1, 2, 4, 8\}$

$\therefore\ A \cap B = \{3, 5\} \cap \{1, 2, 4, 8\}$

$= \emptyset$

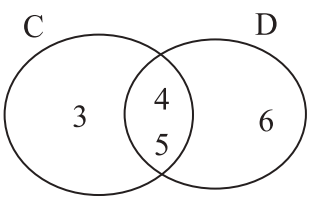
Hence, A and B are disjoint sets.

Example 12. If $C = \{3, 4, 5\}$ and $D = \{4, 5, 6\}$, find $C \cup D$ and $C \cap D$.

Solution : Given that, $C = \{3, 4, 5\}$ and $D = \{4, 5, 6\}$

$\therefore\ C \cup D = \{3, 4, 5\} \cup \{4, 5, 6\} = \{3, 4, 5, 6\}$

and $C \cap D = \{3, 4, 5\} \cap \{4, 5, 6\} = \{4, 5\}$



Activity :
If $P = \{2, 3, 4, 5, 6, 7\}$ and $Q = \{4, 6, 8\}$
1. Find $P \cup Q$ and $P \cap Q$.
2. Express $P \cup Q$ and $P \cap Q$ in set builder form.

Example 13. Express the set $E = \{x : x \text{ is a prime number and } x < 30\}$ in tabular method.

Solution : The required set will be the set of prime numbers less than 30.
Here, the prime numbers less than 30 are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29$

$\therefore$ Required set is $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$.

Example 14. If A and B are sets of all factors of 42 and 70 respectively, find $A \cap B$.

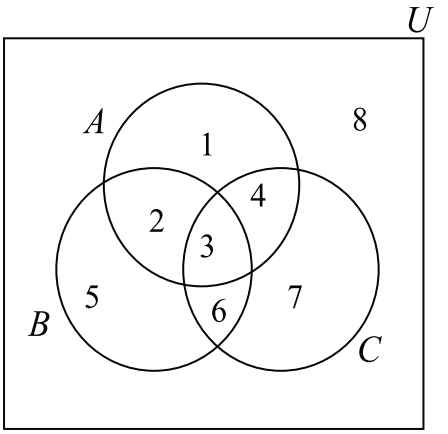
Solution :
Here $42 = 1 \times 42 = 2 \times 21 = 3 \times 14 = 6 \times 7$
Factors of 42 are $1, 2, 3, 6, 7, 14, 21, 42$
$\therefore$ $A = \{1, 2, 3, 6, 7, 14, 21, 42\}$
Again, $70 = 1 \times 70 = 2 \times 35 = 5 \times 14 = 7 \times 10$
Factors of 70 are $1, 2, 5, 7, 10, 14, 35, 70$
$\therefore$ $B = \{1, 2, 5, 7, 10, 14, 35, 70\}$
$\therefore$ $A \cap B = \{1, 2, 7, 14\}$

Exercise 7

1. Which one of the following is the subset of any set?

   A. { 0 } B. { ∅ } C. ∅ D. (∅)

Answer the questions from 2 to 5 in respect of the adjoining Venn diagram.



2. Which one is universal set ?

   (a) A (b) B (c) $A \cup B$ (d) U

3. Which one is the set B^c ?

   (a) $\{5, 6, 7, 8\}$ (b) $\{2, 3, 5, 6\}$ (c) $\{1, 4, 7, 8\}$ (d) $\{3, 6\}$

4. Which one is the set $A \cap B$?

   (a) $\{2, 3\}$ (b) $\{2, 3, 5, 6\}$ (c) $\{3, 4, 6, 7\}$ (d) $\{2, 3, 4, 5, 6, 7\}$

5. Which one is the set $A \cup B$?

   (a) $\{1, 2, 3, 4, 5, 6\}$ (b) $\{5, 6, 7\}$ (c) $\{8\}$ (d) $\{3\}$

6. Express the following sets in tabular form:

   (a) $\{x : x$ is odd number and $3 < x < 15\}$

   (b) $\{x : x$ is a prime factor of $48\}$

   (c) $\{x : x$ is a multiple of 3 and $x < 36\}$

   (d) $\{x : x$ is an integer and $x^2 < 10\}$

7. Express the following sets in set builder form.

   (a) $\{3, 4, 5, 6, 7, 8\}$ (b) $\{4, 8, 12, 16, 20, 24\}$ (c) $\{7, 11, 13, 17\}$

8. If $A = \{1, 2, 3\}$, $B = \{2, a\}$ and $C = \{a, b\}$, find the following sets :

(a) $A \cup B$ (b) $B \cap C$

(c) $A \cap (B \cup C)$ (d) $(A \cup B) \cup C$

(e) $(A \cap B) \cup (B \cap C)$

9. If $U = \{1,2,3,4,5,6,7\}$, $A = \{1,2,5\}$, $B = \{2,4,7\}$ and $C = \{4, 5, 6\}$ justify the correctness of the following relations :

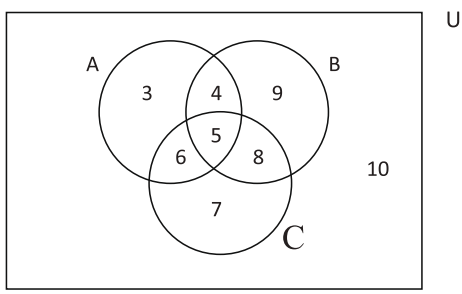
(a) $A \cap B = B \cap A$

(b) $(A \cap B)' = A' \cup B'$

(c) $(A \cup C)' = A' \cap C'$

10. If P and Q are the sets of all the factors of 21 and 35 respectively, find $P \cup Q$.

11.



Figure

A. Write Set A in Set Builder Method.

B. Express A, B and C in Tabular method and find A∩C and A∪B.

C. Prove that (A∪B)′ = A′∩B′.

12. The sets of the natural integers by which the numbers 346 and 556 are divided with remainder 31 in each case are A and B.

A. Express set A in Set Builders Form.

B. Find $A \cap B$.

C. Show $A \cap B$ in Venn-diagram and write the subsets of $A \cap B$.

Sample Questions

Multiple Choice Questions:

1. How many systems are there to express set?
   A. 1 B. 2 C. 3 D. 4

2. S = {x:x is even number and $1 \leq x \leq 7$} which one of the following is correct in Tabular set system?
   A. {2, 3, 4} B. {2, 4, 6} C. {1, 3, 5} D. {3, 5, 7}

3. If, A = {2, 3, 5}
   i. A = {X∈N : 1 <x<6 and x is a prime number}
   ii. A = {X∈N : $2 \leq x < 7$ and x is a prime number}
   iii. A = {X∈N : $2 \leq x \leq 5$ and x is a prime number}

   Which one of the following is correct?
   A. i and ii B. i and iii C. ii and iii D. i, ii and iii

Answer the questions 4 & 5 in light of the information below :

U {2, 3, 5, 7}, A = {2, 5}, B = {3, 5, 7}

4. Which one is A^C?
   A. {2, 5} B. {3, 5} C. {3, 7} D. {2, 7}

5. Which one is $A \cap B^C$?
   A. { 2 } B. { 5 } C. {2, 5} D. 3, 7}

Creative Questions:

6. In a hostel, 65% of the students like fish, 55% of the students like meat and 40% of the students like both dishes.

   (a) Write the subsets of A = { x: x is an even natural number and $x^2 \leq 16$ }.

   (b) Find out the percentage of students who dislike both dishes.

   (c) Find out the intersection set of the sets of factors of those students, who like only one dish.

7. The universal set is $U = \{1,2,3,4,5,6,7\}$. Consider three subsets:,

   $A = \{x: x^2 - 6x + 8 = 0\}$,

   $B = \{x: x$ is an odd number and $1 \leq x \leq 5\}$

   $C = \{x: x$ is a prime number and $2 \leq x \leq 6\}$

   a) Express $P = \{3,5,7,11\}$ in set-builder notation.

   b) Write all the subsets of set A

   c) Find the complement of $(B \cup C)^C$

Short Answer Questions:

8. a) If $A = \{x \in \mathbb{N}: x$ is even, $1 \leq x \leq 6\}$, write all proper subsets of A.

   b) $If\ U = \{1,2,3,4,5,6\}$ and $B = \{2,4,6\}$, find the complement of B^C.

   c) Express $B = \{x \in \mathbb{N}: x$ is a prime factor of 24 $\}$ in tabuiar form.

   d) If $P = \{x: x$, a multiple of 6 $\}$ and $Q = \{x: x$, a multiple of 8$\}$, find $P \cup Q$.